

The University of the South Pacific

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School of Information Technology, Engineering, Mathematics and Physics

CS324: Distributed System

Test 1 – Semester 2, 2024

Learning Outcome: Outline the importance of distributed systems over stand-alone systems

Time Allowed: 50 mins

Total Marks: 30 (10% of final grade)

Name: _____

Student ID: _____

INSTRUCTIONS:

- All questions are compulsory.
- All answers must be written in the spaces provided in this booklet.
- You are allowed to use a silent, non-programmable calculator.
- Show all working.

Question 1: Explain with reasons whether the following scenarios refer to distributed, centralized, or decentralized systems. (4 marks)

Scenario 1: Students in a study group decide to use a peer-to-peer file-sharing network to exchange notes and resources. Each student's computer acts as both a client and a server, sharing files directly with others in the group. There is no central server; instead, each student is responsible for their own files, and the network continues to function even if some students disconnect.

Scenario 2: Imagine a group of students working on a group project using Google Docs. Each student can edit the document simultaneously from different locations. The document is stored and managed by multiple servers in Google's data centers around the world. Even if one server goes down, the others ensure that the document remains accessible and up-to-date for all users.

ANS: Scenario 1 is a decentralized system - because there is no central server; each participant shares files directly with others, making the system more resilient to individual failures.

scenario 2 is distributed system - because the document is managed by multiple servers working together to provide seamless access and editing capabilities.

Question 2: Determine the type of system architecture for the following scenario. Suppose there are three software layers: user interface, application, and database. Draw a diagram to show the placement of layers on tiers. (4 marks)

Scenario: Many banking applications run on an end-user's machine, where the user prepares transactions and such. Once finished, the application contacts the database on the bank's server and uploads the transactions for further processing.

Solution: 2 tiered architecture with database running on server and the remaining layers on client side.

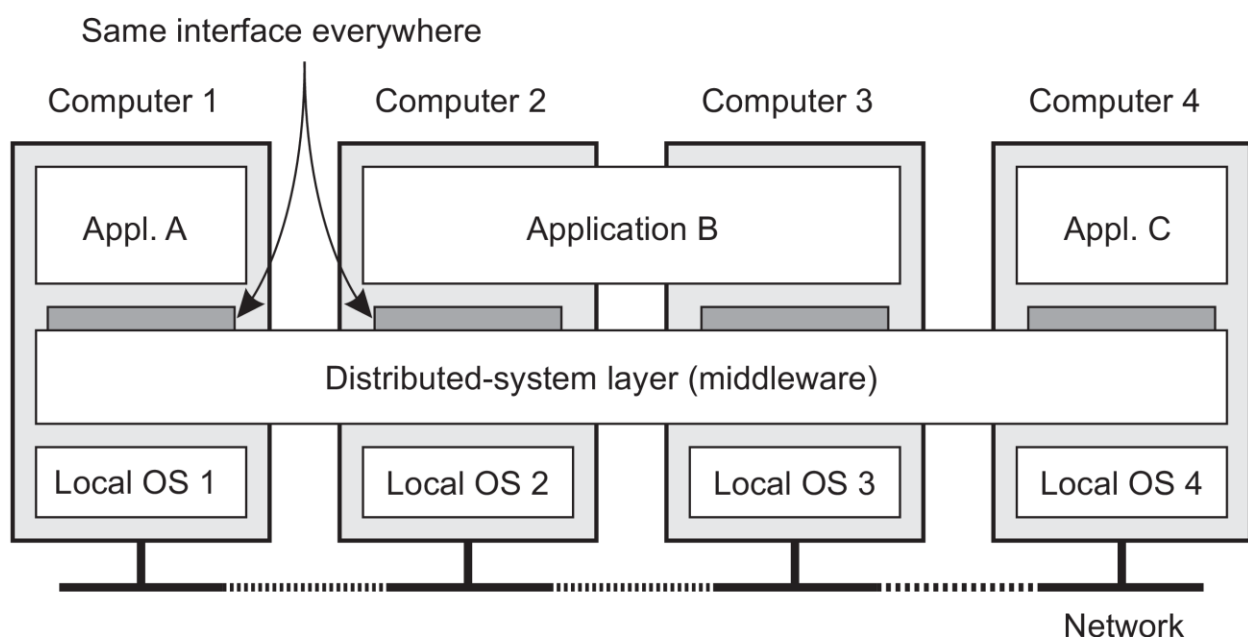
Question 3: You are designing a distributed system for a large-scale sensor network in an industrial setting. Sensors need to share data in real-time for monitoring and control applications. Explain the architecture you would use in this situation? (4 marks)

ANS: I would employ a publish-subscribe messaging pattern. Sensors would publish data to specific topics, and subscribers interested in that data would receive updates in real-time. A distributed message broker or middleware would be used to manage the message routing and delivery. This approach ensures timely and reliable data sharing across the sensor network.

Question 4: Explain an important property of threads that makes it attractive to use in distributed systems. (3 marks)

ANS: An important property of threads is that they can provide a convenient means of allowing blocking system calls without blocking the entire process in which the thread is running (assuming we do not have a many-to-one threading model).

Question 5: Explain the purpose of the distributed-system layer (i.e., middleware) in the context of the figure given below. (4 marks)



ANS: The distributed system provides the means for components of a single distributed application to communicate with each other, but also to let different applications communicate. At the same time, it hides, as best and reasonably as possible, the differences in hardware and operating systems from each application.

It is also a manager of resources and provides services such as security, interprocess communication, etc

Question 6: Explain which distribution transparency is best suited for the following scenario. (3 marks)

Scenario: Online tracking and tracing of goods as they are being transported from one place to another, and teleconferencing (partly) using devices that are equipped with mobile

ANS: Migration transparency - supports the mobility of processes and resources initiated by users, without affecting ongoing communication and operations Internet.

Question 7: Suppose you want to expand a distributed system by adding more machines. Name three scaling techniques that you will apply so that the system does not suffer from scalability problems. (3 marks)

ANS: Hiding communication latency, distribution of work and replication.

Question 8: Name three classifications of distributed systems and provide an example for each classification. (5 marks)

Ans: High-performance distributed computing – cluster, cloud, grid

Distributed information systems – transaction processing, enterprise application integration.

Pervasive systems – Ubiquitous computing, Mobile computing systems